

Q1.

a) Hexadecimal number:

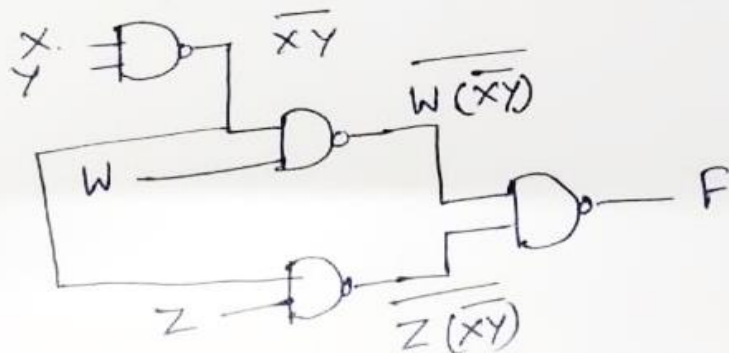
$$\begin{aligned} & (AB.cd)_{16} \\ &= (\overline{010} \overline{010} \overline{11} . \overline{1100} \overline{11010})_{16} \\ &= (253.632)_8 \end{aligned}$$

b) Decimal $\rightarrow 43$

Hexadecimal $\rightarrow (2B)_{16}$ $\begin{array}{r} 16 \overline{)43} \\ \underline{32} \\ 11 \end{array}$

BCD no. $\rightarrow (01000011)_2$

c) $F = W(\overline{xy}) + Z(\overline{xy})$



Ans: 4 2-input NAND gates are required

d) $(0.28C)_{16} \rightarrow (?)_2$

$(0.28C)_{16} \rightarrow \text{~~0.001010001100~~}$
 $(0.001010001100)_2$

$$e) (86)_{16} - (2A)_{16}$$

$$(86)_{16} \rightarrow (10000110)_2$$

$$(2A)_{16} \rightarrow (00101010)_2$$

2's complement of $(2A)_{16} \rightarrow$

$$(11010110)_2$$

$$\begin{array}{r} 10000110 \\ + 11010110 \\ \hline * 01011100 \end{array}$$

$$\text{Ans: } (01011100)_2 \rightarrow (5C)_{16}$$

$$f) (19)_{10} - (23)_{10} = ?$$

$$(19)_{10} \rightarrow (010011)_2$$

$$(23)_{10} \rightarrow (010111)_2$$

$(-23)_{10} \rightarrow$ 2's complement of $(23)_{10}$

$$\rightarrow (101001)_2$$

$$\begin{array}{r} 010011 \\ + 101001 \\ \hline 11100 \end{array} \rightarrow \text{-ve number}$$

$\downarrow$ 2's complement

$$- (00100)_2 \rightarrow -(4)_{10}$$

2. Convert $AB\bar{C} + \bar{A}\bar{B}C + \bar{A}\bar{D} + \bar{B}\bar{C} + AB\bar{D} + ABCD$ into POS using K-map only.

(a) $AB\bar{C} + \bar{A}\bar{B}C + \bar{A}\bar{D} + \bar{B}\bar{C} + AB\bar{D} + ABCD$

$$= AB\bar{C}D + AB\bar{C}\bar{D} + \bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{D}B + \bar{A}\bar{D}\bar{B}$$

$$+ \bar{B}\bar{C}A + \bar{B}\bar{C}\bar{A} + AB\bar{D}C + AB\bar{D}\bar{C} + ABCD$$

$$= AB\bar{C}D + AB\bar{C}\bar{D} + \bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}C\bar{D}$$

$$+ \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D}$$

$$+ \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}\bar{D}$$

$$+ AB\bar{D}C + AB\bar{D}\bar{C} + ABCD$$

= 1101, 1100, 0011, 0010, 0110, 0100, 0010, 0000, 1001, 1000, 0001, 0000, 1110, 1100, 1111

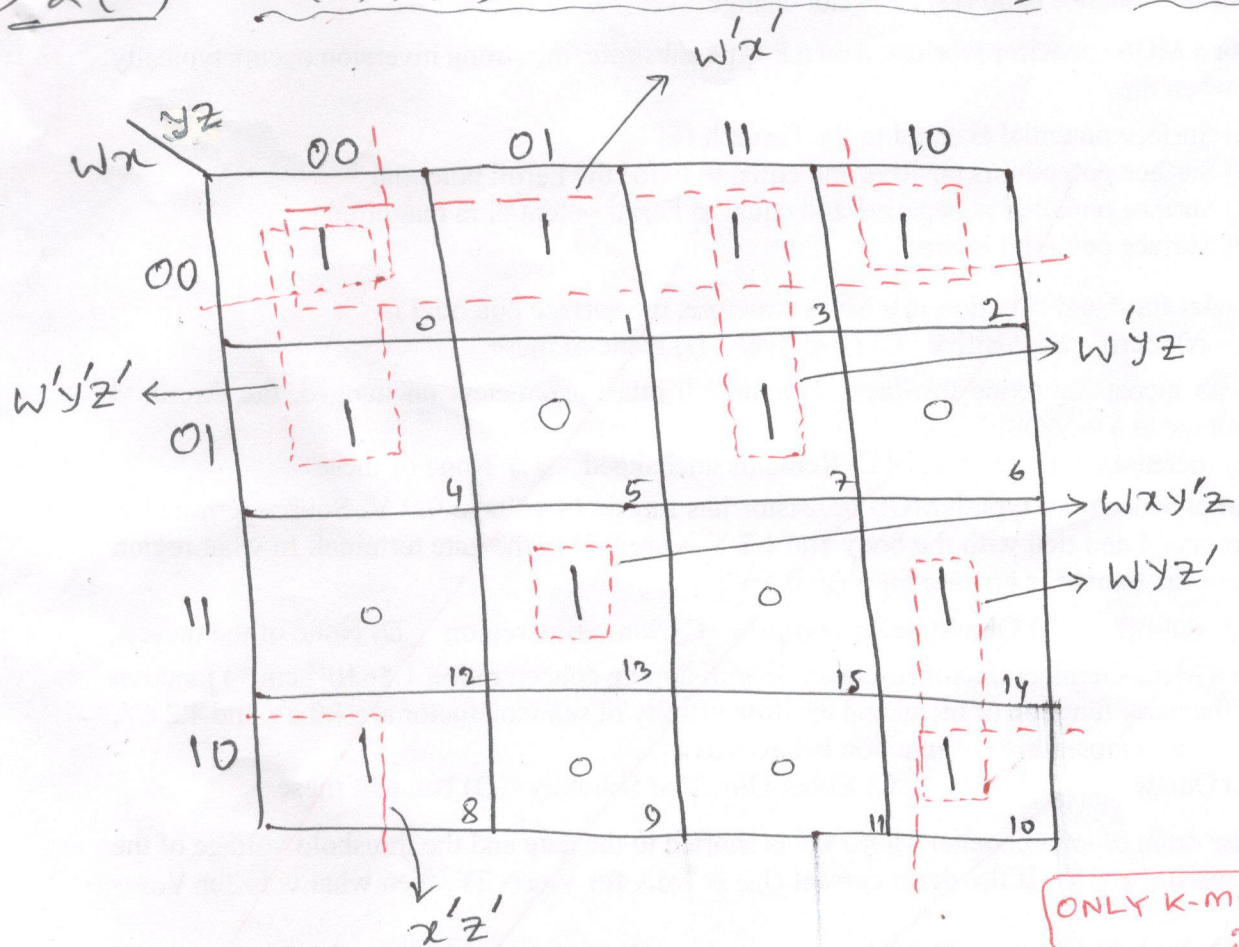
(3)

		00	01	11	10
AB	00	1	1	1	1
	01	1	0	0	1
	11	1	1	1	1
	10	1	1	0	0

POS: $F = (A+B'+D')(\bar{A}+B+C')$

(4)

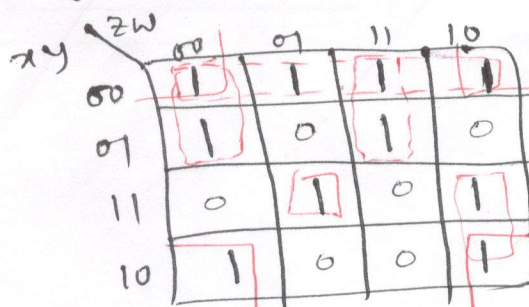
8.2(b) $F(w, x, y, z) = \Pi M(5, 6, 9, 11, 12, 15)$; SOP = ?



$$F = w'x' + x'z' + w'y + wyz' + w'y'z' + wxy'z$$

----- (5) -----

For $F(x, y, z, w) = \Pi M(5, 6, 9, 11, 12, 15)$



$$F = x'y' + y'w' + x'z'w' + x'zw + xzw' + xyz'w$$

For $F(A, B, C, D) = \bar{A}\bar{B} + \bar{B}\bar{D} + \bar{A}\bar{C}\bar{D} + \bar{A}CD + A\bar{C}\bar{D} + ABCD$

Q.3. (a)

$$F = \overline{(B\bar{C} + D)} + D + BC + AD$$

$$= \overline{B\bar{C}} \cdot \bar{D} + D + BC + AD$$

$$= (B + C)\bar{D} + D(1 + A) + BC$$

$$= (B + C)\bar{D} + D + BC$$

$$= (D + \bar{D})(D + B + C) + BC$$

$$= B + C + D + BC$$

$$= B + D + C(1 + B)$$

$$= B + C + D$$

3(b)

$$F(x, y, z) = (xy + z)(xz + y)$$

$$= (x + z)(y + z)(x + y)(z + y)$$

$$= (x + y)(x + z)(y + z) = (x + y + z\bar{z})(x + z + y\bar{y})(x\bar{x} + y + z)$$

$$= (x + y + z)(x + y + \bar{z})$$

$$(x + z + y)(x + z + \bar{y})(x + y + z)$$

$$(x + y + z)$$

$$= (x + y + z)(x + y + \bar{z})(x + \bar{y} + z)$$

$$(x + y + z)$$

$$= \Pi M(0, 1, 2, 4) \rightarrow \text{Product of maxterms}$$

$$= \sum m(3, 5, 6, 7) \rightarrow \text{Sum of minterms}$$

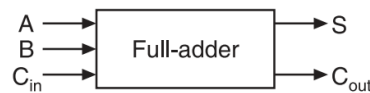
$$= \bar{x}y\bar{z} + x\bar{y}\bar{z} + xy\bar{z} + xyz$$

Question 4 (a): Explain how a full adder can be implemented using half adders showing the necessary derivations and draw the block diagram for Full Adder. (6 Marks)

Solution 4 (a):

Inputs			Sum	Carry
A	B	C_{in}	S	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

(a) Truth table



(b) Block diagram

From the truth table, a circuit that will produce the correct sum and carry bits in response to every possible combination of A, B, and C_{in} is described by

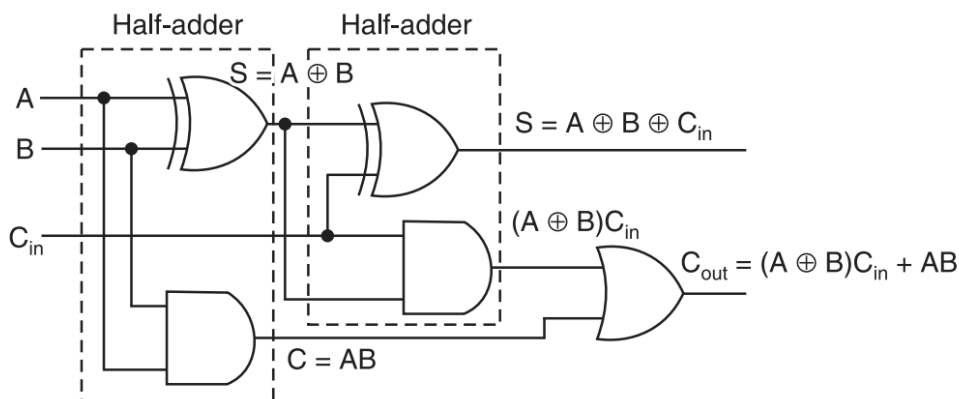
$$S = \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}\bar{C}_{in} + ABC_{in}$$

$$= (\bar{A}\bar{B} + \bar{A}B)C_{in} + (A\bar{B} + AB)\bar{C}_{in} = (A \oplus B)\bar{C}_{in} + (\overline{A \oplus B})C_{in} = A \oplus B \oplus C_{in}$$

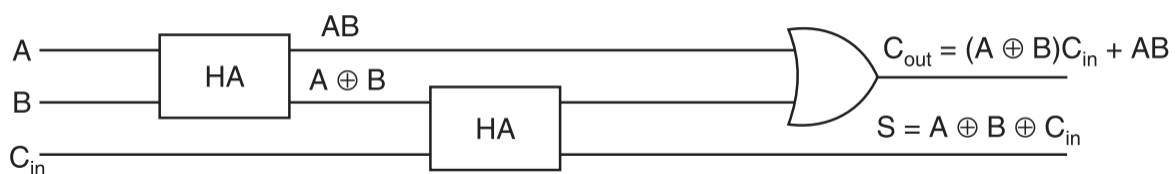
and

$$C_{out} = \bar{A}BC_{in} + A\bar{B}C_{in} + AB\bar{C}_{in} + ABC_{in} = AB + (A \oplus B)C_{in}$$

Logic diagram of a full-adder using two half-adders



Block diagram of a full-adder using two half-adders



Question 4 (b): Find out the Boolean expression for $A > B$ where A and B are two 4-bit binary numbers. Also draw the logic circuit for $A = B$. (3 + 3 Marks)

Solution 4 (b):

The logic for a 4-bit magnitude comparator: Let the two 4-bit numbers be $A = A_3A_2A_1A_0$ and $B = B_3B_2B_1B_0$.

1. If $A_3 = 1$ and $B_3 = 0$, then $A > B$. Or
2. If A_3 and B_3 coincide, and if $A_2 = 1$ and $B_2 = 0$, then $A > B$. Or
3. If A_3 and B_3 coincide, and if A_2 and B_2 coincide, and if $A_1 = 1$ and $B_1 = 0$, then $A > B$. Or
4. If A_3 and B_3 coincide, and if A_2 and B_2 coincide, and if A_1 and B_1 coincide, and if $A_0 = 1$ and $B_0 = 0$, then $A > B$.

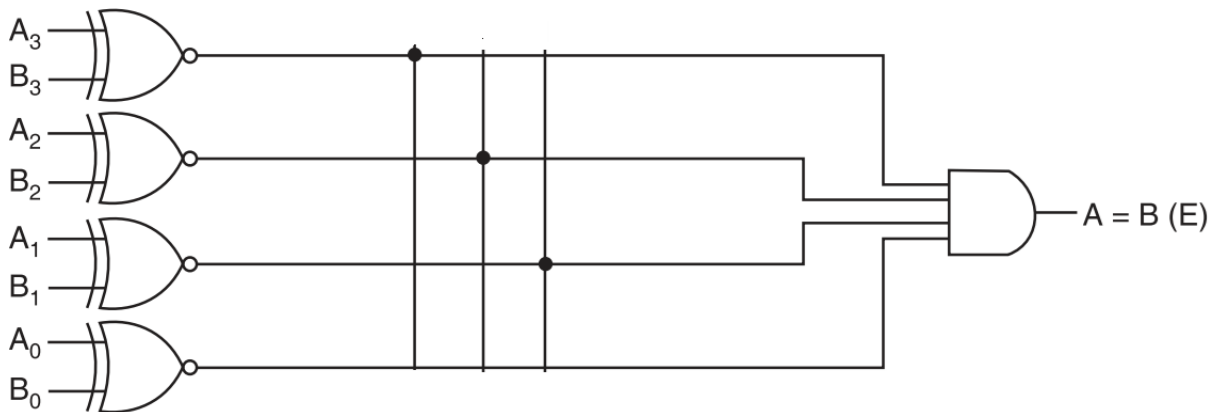
From these statements, we see that the logic expression for $A > B$ can be written as

$$(A > B) : G = A_3\bar{B}_3 + (A_3 \odot B_3)A_2\bar{B}_2 + (A_3 \odot B_3)(A_2 \odot B_2)A_1\bar{B}_1 + (A_3 \odot B_3)(A_2 \odot B_2)(A_1 \odot B_1)A_0\bar{B}_0$$

If A_3 and B_3 coincide and if A_2 and B_2 coincide and if A_1 and B_1 coincide and if A_0 and B_0 coincide, then $A = B$.

So the expression for $A = B$ can be written as

$$(A = B) : E = (A_3 \odot B_3)(A_2 \odot B_2)(A_1 \odot B_1)(A_0 \odot B_0)$$



Question 5

a) How many 4:1 multiplexers are required to design a 32:1 multiplexer (without using other logic gates)?

(2 Marks: show steps-2 marks; direct answer- 1 mark)

$$\begin{array}{l} \text{High order} \rightarrow 32 \rightarrow 8 \\ \text{Lower order} \rightarrow 4 \\ \frac{8}{4} \rightarrow 2 \\ \frac{2}{4} \rightarrow \frac{1}{2} \Rightarrow 1 \end{array}$$

Tot. No. of
 $4:1 \text{ MUX} = 8 + 2 + 1$
 $= 11$

b) Design a 16:1 multiplexer using 8:1 multiplexers and/or necessary logic circuits. (students can design using two 8:1 MUX in 1st stage and in the 2nd stage they can use 8:1 Mux or gates) (4 marks)

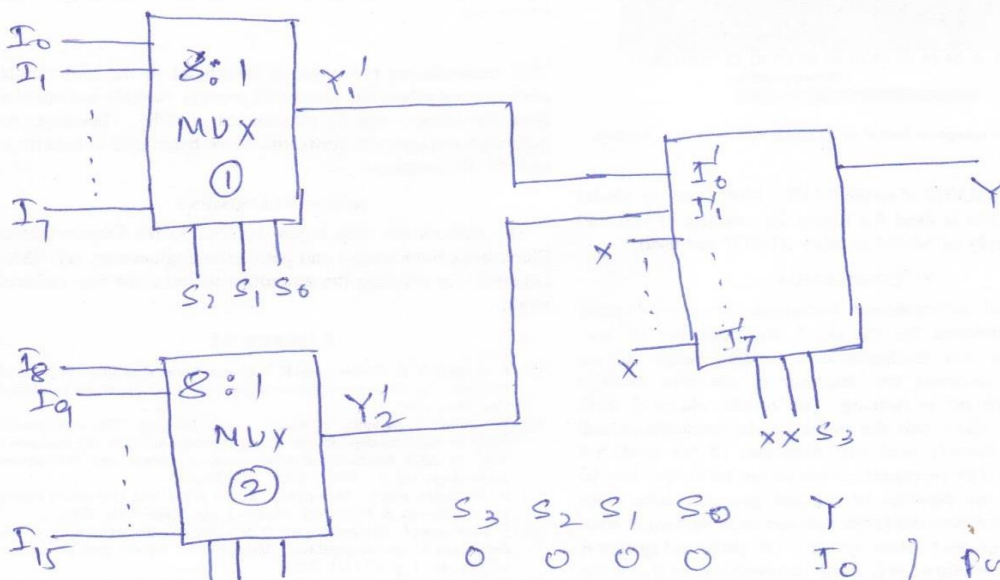
No. of MUX (8:1) required to design a 16:1 MUX

$$= \frac{16}{8} \rightarrow 2$$

$$\text{Tot.} = 2 + 1 = 3$$

$$\frac{2}{8} \rightarrow \frac{1}{4} \Rightarrow 1$$

In the 1st stage 2 no. of 8:1 MUX is required.
 In the 2nd stage, 8:1 MUX can be used or gates can also be used.



S_3	S_2	S_1	S_0	Y	
0	0	0	0	I_0	$\left. \begin{array}{l} \text{For } S_3 = 0; \\ \text{MUX ① will} \\ \text{be selected} \end{array} \right\}$
0	0	0	1	I_1	
0	0	1	0	I_2	
0	0	1	1	I_3	
1	0	0	0	I_8	$\left. \begin{array}{l} \text{For } S_3 = 1 \\ \text{MUX ② will} \\ \text{be selected} \end{array} \right\}$
1	0	0	1	I_9	
1	0	1	0	I_{10}	
1	0	1	1	I_{11}	

